Applied Social Data Science

Coding Camp

Setup · Orientation

Elena Karagianni
karagiae@tcd.ie

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Department of Political Science, Trinity College Dublin

Schedule

10am - 1pm

- Monday: Introduction + R Basics I
- **Tuesday: R Basics II + Good practices**

10am - 12pm

- Thursday: Python basics
- Friday: How to write up and report on LaTeX

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- Learn by doing: code is a language, and it's always easier to learn a language by using it – so find some data that interests you, and put your skills to work.
- Take advantage of online learning resources: sites like Datacamp, Udemy, and Kaggle can help improve your data science skills. Online free-access books too!

Good practices for doing data science

Organize your files!

Use folders to organize your projects. Set a working directory for each project and keep all related files inside it.

R / RStudio

- Set directory: `setwd("path")`
- Check directory: `getwd()`
- Use **Projects** in RStudio to assign directories automatically

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Python

- Import package: `import os`
- Change directory: `os.chdir("path")`

Work *with* your IDE

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In general...

Work **with** your IDE, not **for** it. Relying too much on automatic features (e.g. auto-restore workspace) can make your scripts unreliable.

The importance of a good script

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A good script...

- begins with the required packages/libraries/modules
- is well-commented to explain the workflow
- is saved in the project's working directory and points there when creating/saving files
- uses meaningful names for objects

Naming objects

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In any data science project you will create numerous objects: datasets, variables, functions, etc. Consistent naming makes it easier for **you** to revisit old scripts and for **others** to understand your work.

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Remember:

Not all case styles will work in all languages. For example, *`dont.do.this.in.python`*